

Iris Flower Classification

Project Overview

This project is part of a machine learning internship task.

The goal is to classify Iris flowers into three species: Setosa, Versicolor, and Virginica based on their measurements.

Objective

To build a machine learning model that can:

- Learn from Iris dataset features
- Classify flowers into correct species
- Achieve high prediction accuracy

Dataset

The Iris dataset contains 150 samples with 4 features:

Features:

- Sepal Length (cm)
- Sepal Width (cm)
- Petal Length (cm)
- Petal Width (cm)

Target:

- Species (Setosa, Versicolor, Virginica)

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Model Used

- Logistic Regression

(You can also try Decision Tree, KNN, Random Forest)

Results

- Model Accuracy: ~100% (on test data)

- Perfect classification for all three species

Workflow

1. Load dataset
2. Data exploration
3. Data visualization
4. Train-test split
5. Model training
6. Prediction
7. Evaluation

How to Run

1. Open Jupyter Notebook
2. Install required libraries:
pip install pandas numpy matplotlib seaborn scikit-learn
3. Run all cells step by step

Project Structure

Iris-Flower-Classification/

Iris_Flower_Classification.ipynb

README.md

Conclusion

This project demonstrates a basic machine learning classification model using the Iris dataset with excellent accuracy.

Author

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